

# MAHINDRA & MAHINDRA

## SERVICE QUALITY & CUSTOMER WAITING-TIME REDUCTION

### Executive Context & Scope:

This case study presents an independent Quality Engineering analysis of customer waiting-time management and process optimization within an automotive service environment. By combining Total Quality Management (TQM) principles, detailed customer journey mapping, root-cause diagnostics (5 Why & Shainin Red X), and structured PDCA execution, this report outlines a practical framework for reducing operational delays while enhancing overall customer satisfaction.

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### 1. PROBLEM IDENTIFICATION: WAITING TIME ACROSS THE SERVICE JOURNEY

Customer waiting time in an automotive service centre is rarely created by a single problem. A delay may begin while getting an appointment, during workshop check-in, while waiting for the vehicle to be attended, because of uneven workload, lack of required parts or resources, or because the expected completion time is not communicated clearly. A delay at one stage can also affect the customers and activities that follow.

For this reason, waiting time should not be treated only as a scheduling problem. It should be studied as part of the complete service process. The objective is to understand where delays are being created, how they move through the service journey, and how they finally affect the customer's overall service experience.

For this case, the focus is not only on making the service faster. The aim is to reduce customer waiting time while improving the overall service experience, without disturbing the parts of the process that are already working effectively.

### 2. UNDERSTANDING THE COMPLETE SERVICE PROCESS

Improvement should begin by understanding the process before attempting to change it. The complete customer journey can be mapped from the first service appointment request to the final feedback and follow-up. This makes it easier to see where customers are waiting, where work is getting delayed, and where coordination between different service activities may be creating a gap.

The proposed service flow begins with Service Appointment Request, followed by Appointment Confirmation, Customer Arrival at Workshop, Vehicle Reception and Check-in, Initial Vehicle Inspection, Service Requirement Confirmation, Job Card Creation, Parts and Material Availability Check, Technician Assignment, Service or Repair Work, Quality Check, Final Vehicle Check, Service Completion, Billing and Payment, Customer Notification, Vehicle Handover, Customer Feedback, Complaint Resolution if required, and finally Feedback Analysis and Continuous Improvement.

Process mapping alone, however, is not enough. Waiting-time observations should be combined with actual process data and customer feedback. Together, these three inputs help distinguish an assumed problem from the point where the customer is actually experiencing delay.

### 3. MANAGING THE SERVICE QUEUE: FIFO WITH DEFINED PRIORITY RULES

Once a vehicle has completed workshop check-in, First-In, First-Out (FIFO) can be used as the basic principle for managing the normal service queue. Vehicles that enter the applicable service queue earlier should normally be attended before those entering later. This provides a simple and visible sequence instead of allowing the order of service to change without a clear reason.

FIFO should not, however, be applied blindly. Customer problems are not always equal in urgency or impact. A defined exception rule can therefore be used for genuine priority situations. Instead of creating a completely separate priority queue, the normal FIFO sequence can remain in place while specific cases are given priority according to clearly defined conditions.

The main performance measure in this case is customer waiting time. The purpose of queue management is not simply to move more vehicles through the workshop, but to reduce the time and uncertainty experienced by the customer throughout the service journey.

### 4. FINDING THE CAUSE: 5 WHY AND SHAININ RED X

After the process has been mapped and the major delay points have been identified, the next step is to understand why those delays are occurring. 5 Why Analysis can be applied to each important problem by repeatedly asking why until the underlying cause becomes clearer. A delay that initially appears to be caused by a technician, for example, may eventually be connected to workload allocation, information flow, parts availability, training or another earlier stage of the process.

Several possible causes may emerge from this analysis. Instead of trying to change everything at the same time, the Shainin Red X approach can be used to search for the dominant source contributing most strongly to the observed variation or problem. This helps focus improvement effort on the cause that has the greatest influence rather than spreading resources across every possible factor.

The purpose is not to assume beforehand that only one cause exists. The process first identifies the possible causes and then uses evidence to determine which cause or causes deserve the highest attention.

### 5. CORRECTIVE ACTION THROUGH THE PDCA CYCLE

Once the important cause has been identified, a suitable corrective action can be tested. The action should directly address the identified problem rather than changing unrelated parts of the service process. If a delay is linked to lack of support or training, for example, the response should focus on that gap. If it is linked to scheduling, parts availability or coordination, the corrective action should be designed around that specific issue.

The overall improvement can be managed through the PDCA cycle. In the Plan stage, the problem, data and proposed corrective action are defined. During Do, the improvement is tested on a controlled basis. Check compares the result with the earlier service performance, and Act determines whether the change should be continued, adjusted or standardized.

An important principle throughout this cycle is to improve the identified weak area without unnecessarily changing activities that are already performing well.

### 6. VALIDATING THE IMPROVEMENT: PROCESS DATA + CUSTOMER FEEDBACK

A reduction in recorded waiting time alone does not prove that the complete service experience has improved. The result should therefore be evaluated using both before-and-after customer waiting-time data and direct customer feedback.

Customer feedback creates an important feedback loop because the improvement is ultimately being made for the customer. It can indicate whether the customer actually experienced shorter waiting, clearer communication and a better overall service journey. At the same time, process data provides objective evidence of whether waiting time has genuinely changed.

If waiting time improves but customer feedback remains poor, the successful improvement should not automatically be discarded. Instead, the remaining source of dissatisfaction should be identified and studied separately, while preserving the parts of the process that are already working well. This keeps improvement focused rather than repeatedly changing the entire system.

### 7. STANDARDIZATION & CONTINUOUS IMPROVEMENT

A successful trial should not immediately become a permanent standard. Once the improved method shows positive results, it can first be extended to the relevant service teams and monitored over a suitable period. The purpose is to confirm that the improvement remains stable and continues to deliver both lower waiting time and a better customer experience.

If the results remain consistent, the improved method can then be converted into a formal Standard Operating Procedure (SOP). Customer feedback and process performance should continue to be reviewed even after standardization, allowing the next PDCA cycle to begin whenever a new gap appears.

In this way, the service process does not end with one corrective action. It becomes a continuous loop of understanding, improvement, validation and standardization.

### 8. TQM TOOLS USED IN THIS CASE

**Process Mapping:** Process Mapping is a structured representation of the steps involved in a process. In this case, it is used to understand the complete service journey from appointment request to vehicle handover and customer feedback, helping identify where waiting or coordination gaps may occur.

**FIFO (First-In, First-Out):** FIFO is a sequencing principle in which the item or case entering a queue first is normally processed first. In this case, FIFO is used after workshop check-in as the basic service-queue rule, supported by clearly defined exceptions for genuine priority situations.

**5 Why Analysis:** The 5 Why method is a simple root-cause analysis technique in which repeated “Why?” questions are used to move beyond the visible problem and understand its underlying cause. Here, it is used to investigate the major delay points identified during service-process analysis.

**Shainin Red X:** The Red X approach focuses on identifying the dominant source of variation that has the strongest effect on a problem. In this case, it helps prioritize the most influential cause when several possible reasons for service delay have been identified.

**PDCA Cycle:** PDCA stands for Plan, Do, Check and Act. It provides the overall improvement cycle for this case: understand and plan the improvement, test it, evaluate the result, and then either refine or standardize the successful change.

**Customer Feedback Loop:** A customer feedback loop uses customer responses after an improvement to understand whether the change produced the intended experience. Here, customer feedback is combined with actual waiting-time data to validate whether the service improvement worked from both the process and customer perspectives.

**Standardization / SOP:** Standardization converts a proven and stable improvement into a consistent working method. In this case, the improved process is monitored first and, only after sustainable results are confirmed, converted into a formal SOP.

*“From waiting time to customer experience, quality improvement begins with understanding the process.”*

**EDUCATIONAL DISCLAIMER:** This case study has been independently developed for educational and learning purposes as part of the #100DaysOfTQM initiative. It applies TQM and Quality Engineering concepts to a Mahindra & Mahindra service-quality context using independent analysis and publicly available context. It does not represent an official Mahindra & Mahindra project, internal company process or company-approved solution. The author is not representing or claiming affiliation with Mahindra & Mahindra through this case study.